

SpatialLens Assist

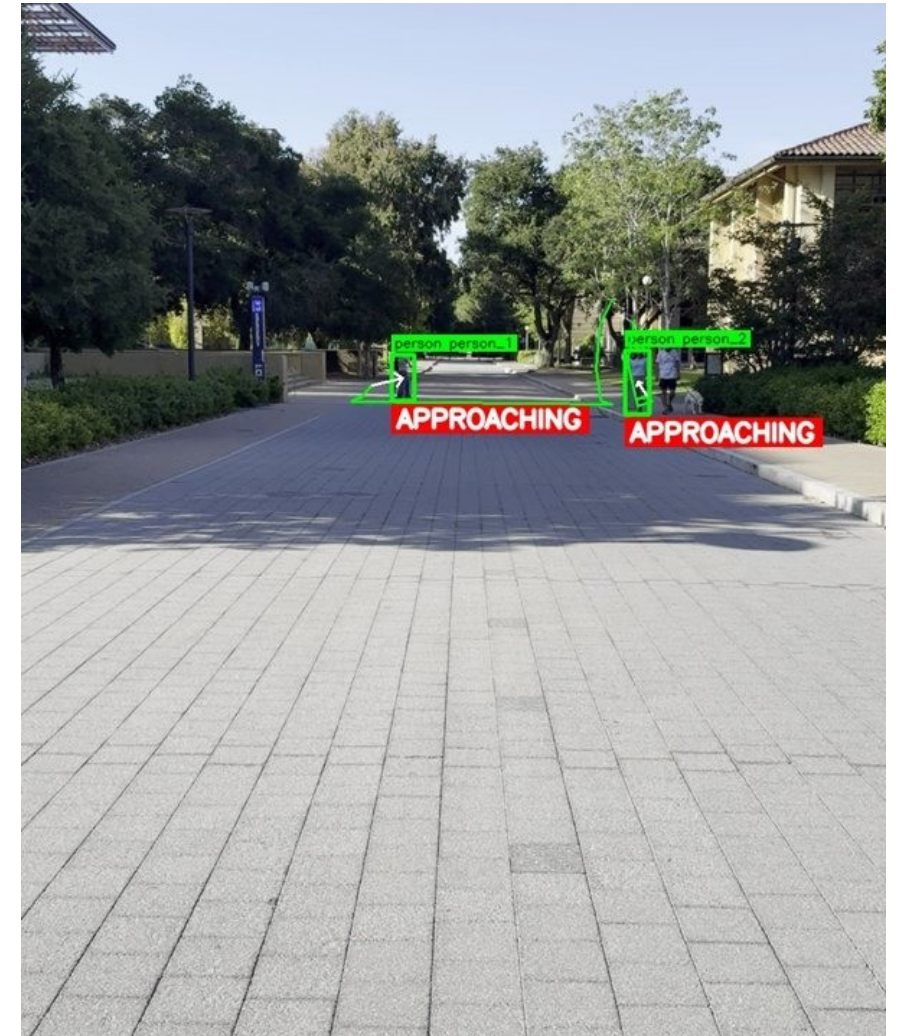
Motion-Aware Hazard Detection for Campus
Navigation
from Low-FPS Egocentric Video

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Detection tells you what, not what it's doing

- A parked bicycle and one bearing down on you are both just “bicycle, 0.86”.
- A sighted person reads intent from context — a blind / low-vision
 - pedestrian can't. The safety signal is motion, not the label.
- **Our task: per-track motion classification from a ~10 s clip @ 2 fps**
 - approaching · crossing_LTR · crossingRTL
 - moving_away · static · uncertain



Hazard overlay on a campus walkway clip

A transparent, CPU-only pipeline



Ego-motion compensation (median flow + ECC) · Evidence string on every hazard label

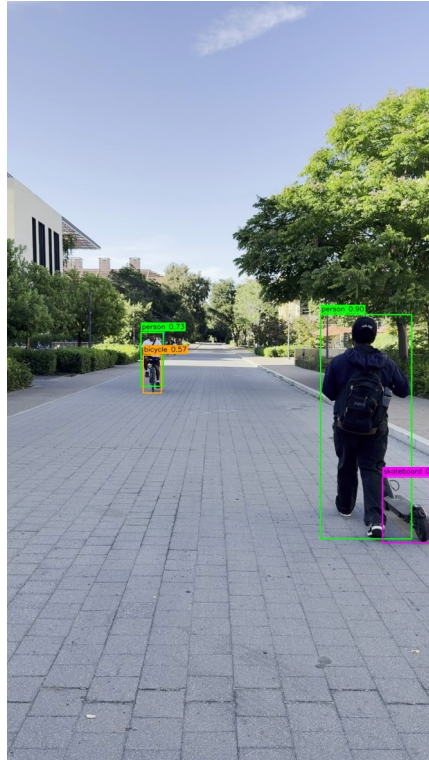
2 fps frames → YOLOv8n → tracker + appearance re-ID → ego-motion-compensated motion features → rule cascade → alert + evidence string

- Explainable — every prediction carries a natural-language evidence string.
- Calibrated abstention — “uncertain” is first-class; short tracks don’t bluff.
- CPU-only — the full pipeline runs in ~4 min on a GPU-less Mac.
- **Key trick: ego-motion compensation cancels camera pan, so static objects stay static.**

From pixels to a spoken alert



1 • Raw frame (2 fps)



2 • YOLOv8n detections



3 • Tracking + re-ID



4 • Hazard overlay



“Bicycle approaching directly ahead.” [high]

evidence string: the cyclist's bounding box grew ~22x; centroid moved toward image center

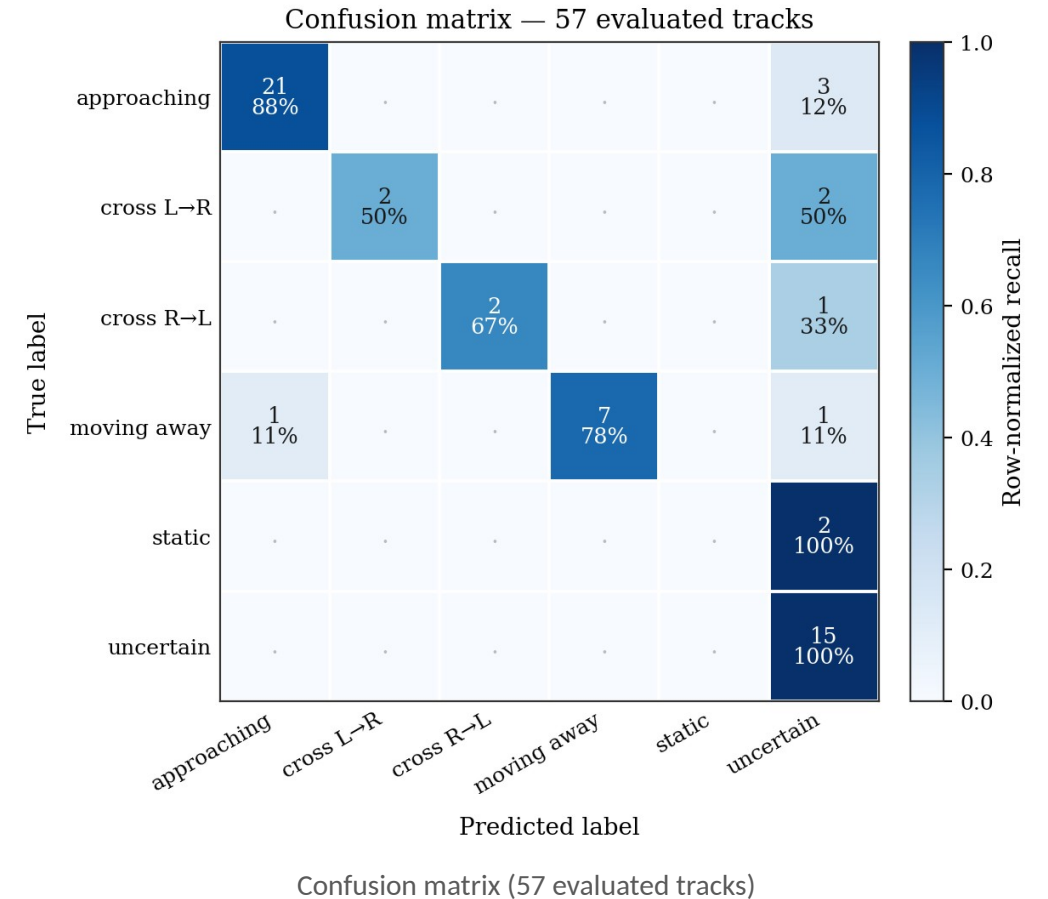
Accurate where it counts

Overall accuracy: **82.5%**

Decidable (≥ 3 -frame tracks): **93.3%**

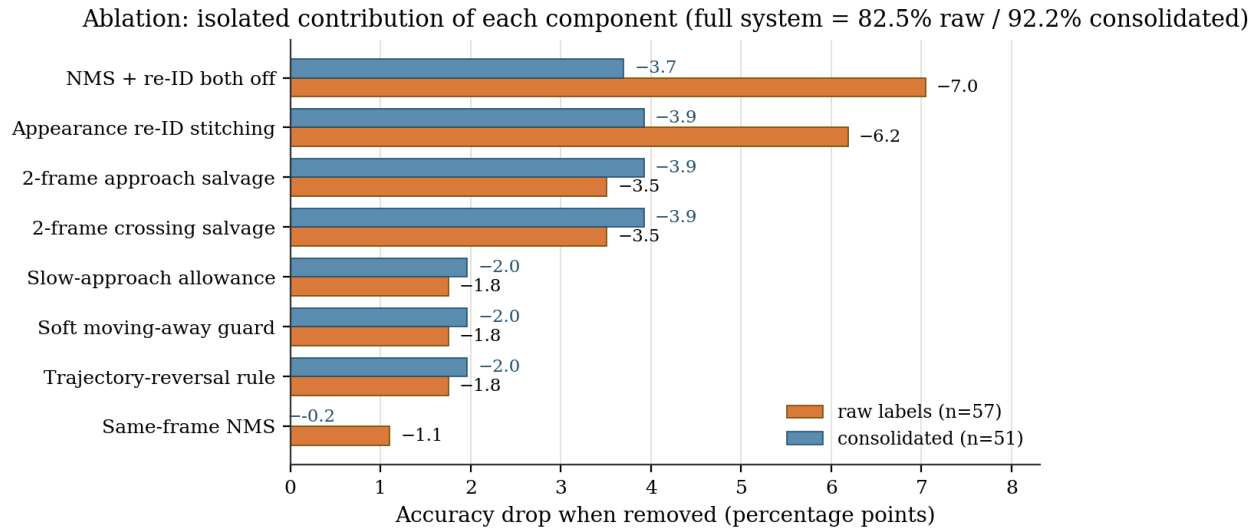
approaching P / R: **0.95 / 0.88**

11 videos • 61 hand-labelled tracks



Zero approaching ↔ moving_away confusion

Continuity beats classifier complexity



Accuracy drop when each component is removed

- **Biggest gains = keeping tracks intact, not fancier rules:**

- appearance re-ID: +6.2 pts (largest single gain)

- two short-track salvage rules: +3.5 pts each

- **Negative result: ByteTrack → 32.8% on 2 fps offline footage —**

- its online motion-continuity assumption breaks under sparse sampling.

- **Abstention = the “fragmentation tax made auditable”:**

- 23 of 24 abstentions are short tracks, not classifier confusion.

Takeaways & what's next

- Simple, interpretable motion cues recover useful hazard labels from cheap
- low-fps video — and continuity matters more than classifier sophistication.
- Next:
 - monocular depth (MiDaS) · learned re-ID embedding
 - denser sampling (5–10 fps) · user study with blind / low-vision walkers

Thank you!